MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, MOS,
4096 BIT STATIC RANDOM ACCESS MEMORY (RAM),
MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, N-channel static MOS, 4096 bit random access memories. Three product assurance classes and a choice of lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit	Address access time
01 (T_C = -55°C "instant-on" to +125°C) 1/	4096 words/1-bit	t_{AA} = 85 ns
02 (T_C = -55°C "instant-on" to +125°C) 1/	1024 words/4-bit	t_{AA} = 450 ns
03 (T_C = -55°C "instant-on" to +125°C) 1/	4096 words/1-bit	t_{AA} = 70 ns
04 (T_C = -55°C "instant-on" to +125°C) 1/	1024 words/4-bit	t_{AA} = 250 ns
05 (T_C = -55°C "instant-on" to +125°C) 1/	4096 words/1-bit	t_{AA} = 55 ns
06 (T_C = -55°C "instant-on" to +125°C) 1/	1024 words/4-bit	t_{AA} = 70 ns
07 (T_C = -55°C "instant-on" to +125°C) 1/	4096 words/1-bit	t_{AA} = 45 ns

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
V	D-6 (18-lead, 1/4" x 15/16", dual-in-line pack)

1/ T_C = T_A at test time equals zero. "Instant-on" is defined as all functional characteristics guaranteed at all temperatures 50 ms after power is applied.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Voltage on any pin with respect to ground (01, 02)	-1.5 to +7.0 V 2/
Voltage on any pin with respect to ground (03, 04, 05, 06, 07)	-3.5 to +7.0 V
Storage temperature range	-65°C to 150°C
Power dissipation:	
Device types 01, 03, 05, 06, 07	1.2 W
Device types 02, 04	1.0 W
Lead temperature (soldering 5 seconds)	270°C
Maximum junction temperature	T_J = 150°C 3/
Thermal resistance junction to case	θ_{JC} = 25°C/W
Maximum DC output current:	
Device types 01, 03, 05, 06, 07	20 mA
Device types 02, 04	5 mA

1.4 Recommended operating conditions.

Device types 01, 02, 03, 04, 05, 06, and 07	Min	Max	Units
Supply voltages:			
V_{CC}	4.5	5.5	Vdc
V_{SS}	0		Vdc
High-level input voltage	2.0	V_{CC}	Vdc
Low-level input voltage	-1.0	+0.8	Vdc
Case temperature	-55°	+125°	C
Device type 01			
Read cycle time (t_{RC})	85		ns
Address access time (t_{AA})		85	ns
CS access time (t_{ACS1}) 4/		85	ns
Chip select access time (t_{ACS2}) 5/		100	ns
Output hold time from address change (t_{OH})	5		ns

2/ Under absolute maximum ratings, the voltage values are with respect to the most negative supply voltage, V_{SS}. Throughout the remainder of this specification, the voltage values are with respect to V_{SS}.

3/ Maximum junction temperature (T_J) may be increased to 175°C during the burn-in and steady state life test.

4/ Chip deselected for greater than 55 ns prior to selection.

5/ Chip deselected for a finite time that is less than 55 ns prior to selection.

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1.4 Recommended operating conditions - Continued.

Device type 01 - Continued.	Min	Max	Units
Chip select to output in low-Z (t_{LZ}) 6/ 7/	10		ns
Chip deselect to output in high-Z (t_{HZ}) 6/ 7/	0	40	ns
Chip select to power-up time (t_{PU})	0		ns
Chip select to power-down time (t_{PD})		30	ns
Write cycle time (t_{WC})	85		ns
Pulse width, chip select to end of write (t_{CW})	70		ns
Address valid to end of write (t_{AW})	70		ns
Pulse width, write (t_{WP})	55		ns
Data valid to end of write (t_{DW})	35		ns
Address set-up time (t_{AS})	0		ns
Write recovery time (t_{WR})	15		ns
Data on hold time (t_{DH})	10		ns
Write enabled to output in high-Z (t_{WZ}) 7/	0	65	ns
Output active from end of write (t_{OW}) 7/	0		ns
Device type 02			
Read cycle time (t_{RC})	450		ns
Address access time (t_{AA})		450	ns
Chip select access time (t_{ACS})		120	ns
Output hold time from address change (t_{OH})	50		ns
Chip select to output in low-Z (t_{LZ}) 6/ 7/	20		ns
Chip deselect to output in high-Z (t_{HZ}) 6/ 7/	0	100	ns
Write cycle time (t_{WC})	450		ns
Pulse width, write (t_{WP})	200		ns
Data valid to end of write (t_{DW})	200		ns
Write recovery time (t_{WR})	0		ns
Data on hold time (t_{DH})	0		ns
Write enabled to output in high-Z (t_{WZ}) 7/	0	100	ns
Device type 03			
Read cycle time (t_{RC})	70		ns
Address access time (t_{AA})		70	ns

6/ At any given temperature and voltage condition, t_{HZ} maximum is less than t_{LZ} minimum both for a given device and from device to device.

7/ Transition is measured ±500 mV from steady state voltage with specified loading.

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1.4 Recommended operating conditions - Continued.

Device type 03 - Continued.	Min	Max	Units
Chip select access time (t_{ACS1}) 8/		70	ns
Chip select access time (t_{ACS2}) 5/		80	ns
Output hold time from address change (t_{OH})	5		ns
Chip select to output in low-Z (t_{LZ}) 6/ 7/	10		ns
Chip deselect to output in high-Z (t_{HZ}) 6/ 7/	0	40	ns
Chip select to power-up time (t_{PU})	0		ns
Chip select to power-down time (t_{PD})		30	ns
Write cycle time (t_{WC})	70		ns
Pulse width, chip select to end of write (t_{CW})	55		ns
Address valid to end of write (t_{AW})	55		ns
Pulse width, write (t_{WP})	40		ns
Data valid to end of write (t_{DW})	30		ns
Address set-up time (t_{AS})	0		ns
Write recovery time (t_{WR})	15		ns
Data on hold time (t_{DH})	10		ns
Write enabled to output in high-Z (t_{WZ}) 7/	0	35	ns
Output active from end of write (t_{OW}) 7/	0		ns
Device type 04			
Read cycle time (t_{RC})	250		ns
Address access time (t_{AA})		250	ns
Chip select access time (t_{ACS})		85	ns
Output hold time from address change (t_{OH})	15		ns
Chip select to output in low-Z (t_{LZ}) 6/ 7/	10		ns
Chip deselect to output in high-Z (t_{HZ}) 6/ 7/	0	60	ns
Write cycle time (t_{WC})	250		ns
Pulse width, write (t_{WP})	135		ns
Data valid to end of write (t_{DW})	135		ns
Write recovery time (t_{WR})	0		ns
Data on hold time (t_{DH})	0		ns
Write enabled to output in high-Z (t_{WZ}) 7/	0	60	ns

8/ Chip selected for greater than 55 ns prior to selection.

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1.4 Recommended operating conditions - Continued.

Device type 05			
Read cycle time (t_{RC})	55		ns
Address access time (t_{AA})		55	ns
Chip select access time (t_{ACS1}) 8/		55	ns
Chip select access time (t_{ACS2}) 5/		65	ns
Output hold time from address change (t_{OH})	5		ns
Chip select to output in low-Z (t_{LZ}) 6/ 7/	10		ns
Chip deselect to output in high-Z (t_{HZ})	0	30	ns
Chip select to power-up time (t_{PU})	0		ns
Chip select to power-down time (t_{PD})		20	ns
Write cycle time (t_{WC})	55		ns
Pulse width, chip select to end of write (t_{CW})	45		ns
Address valid to end of write (t_{AW})	45		ns
Pulse width, write (t_{WP})	25		ns
Data valid to end of write (t_{DW})	25		ns
Address set-up time (t_{AS})	0		ns
Write recovery time (t_{WR})	10		ns
Data on hold time (t_{DH})	10		ns
Write enabled to output in high-Z (t_{WZ}) 7/	0	25	ns
Output active from end of write (t_{OW}) 7/	0		ns
Device type 06			
Read cycle time (t_{RC})	70		ns
Address access time (t_{AA})		70	ns
Chip select access time (t_{ACS1}) 8/		70	ns
Chip select access time (t_{ACS2}) 5/		80	ns
Output hold time from address change (t_{OH})	5		ns
Chip select to output in low-Z (t_{LZ}) 6/ 7/	20		ns
Chip deselect to output in high-Z (t_{HZ}) 6/ 7/	0	20	ns
Chip select to power-up time (t_{PU})	0		ns
Chip select to power-down time (t_{PD})		30	ns
Write cycle time (t_{WC})	70		ns
Pulse width, chip select to end of write (t_{CW})	65		ns
Address valid to end of write (t_{AW})	65		ns

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1.4 Recommended operating conditions - Continued.

Device type 06 - Continued.	Min	Max	Units
Address set-up time (t_{AS})	0		ns
Pulse width, write (t_{WP})	50		ns
Write recovery time (t_{WR})	5		ns
Data valid to end of write (t_{DW})	25		ns
Data on hold time (t_{DH})	0		ns
Write enabled to output in high-Z (t_{WZ}) 7/	0	25	ns
Output active from end of write (t_{OW}) 7/	0		ns
Device type 07			
Read cycle time (t_{RC})	45		ns
Address access time (t_{AA})		45	ns
Chip select access time (t_{ACS1}) 8/		45	ns
Chip select access time (t_{ACS2}) 5/		55	ns
Output hold time from address change (t_{OH})	5		ns
Chip select to output in low-Z (t_{LZ}) 6/ 7/	10		ns
Chip deselect to output in high-Z (t_{HZ}) 6/ 7/	0	30	ns
Chip select to power-up time (t_{PU})	0		ns
Chip select to power-down time (t_{PD})		20	ns
Write cycle time (t_{WC})	45		ns
Pulse width, chip select to end of write (t_{CW})	45		ns
Address valid to end of write (t_{AW})	45		ns
Pulse width, write (t_{WP})	30		ns
Data valid to end of write (t_{DW})	30		ns
Address set-up time (t_{AS})	0		ns
Write recovery time (t_{WR})	10		ns
Data on hold time (t_{DH})	10		ns
Write enable to output in high-Z (t_{WZ}) 7/	0	25	ns
Output active from end of write (t_{OW}) 7/	0		ns

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

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STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and as specified herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Block diagrams. The block diagrams shall be as specified on figure 2.

3.2.3 Functional tests. The functional tests used to test this device are contained in the appendix. If the test patterns cannot be implemented due to test equipment limitations, alternate test patterns to accomplish the same results shall be submitted to the procuring activity for approval.

3.2.4 Truth table. The truth table shall be as specified on figure 3.

3.2.5 Case outline. The case outline shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements shall be as specified in table III. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 46 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. When the alternate screening option of 3.3 of method 5004 is applied to class B and C devices under this specification, the following additional items are applicable:

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1. Internal visual, method 2010, condition B: In addition to the changes indicated by 3.3.1 of method 5004, the following additional clarifications and deletions to method 2010 are applicable:

   (a) Top layer conductors or top polysilicon conductors shall be inspected using the metallization criteria of MIL-STD-883, method 2010. Underlying conductors such as polysilicon need not be inspected.

   (b) Omit 3.2.1.1b through 3.2.1.1e, 3.2.1.2b through 3.2.1.2e, and 3.2.3e (items 3.2.1.1f and 3.2.3g do not apply).

   (c) All gate oxide contacts shall be treated under the criteria of 3.2.1 with the following paragraphs deleted; 3.2.1.1e, 3.2.1.2c, 3.2.1.2d, 3.2.1.2e, 3.2.1.7d, 3.2.1.7e, 3.2.1.1f, and 3.2.3.

2. A cell bit stress test may be used for the special electrical screen test of 3.3.1 of method 5004. The cell bit stress test shall be conducted per table III test number 40 for device types 01, 03, 05, and 07 and test number 50 for device types 02, 04, and 06. The functional test shall be the checkerboard/checkerboard algorithm of the appendix. This test shall be performed once at the first high temperature (125°C) functional test only.

b. Burn-in test (method 1015 of MIL-STD-883).

   1. Test condition D or E, using the circuit shown on figure 4, or equivalent.

c. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameter tests prior to burn-in are optional at the discretion of the manufacturer.

d. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 and 7 tests after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures of group A, subgroups 1 and 7, after burn-in divided by the total number of devices submitted to burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for group A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.3.1 Qualification extension. For qualification inspection, if a manufacturer qualifies to a faster device type which is designed and manufactured identically to a slower device type on this specification, then the slower device type may be part I qualified by conducting only group A electrical tests (when authorized by the qualifying activity) and submitting data in accordance with MIL-M-38510, appendix D (i.e. group B, C and D tests are not required).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for group A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ 2/ 5/	Device type	Limits Max	Limits Min	Unit
Low level input leakage current (all input pins)	I_{IL}	V_{CC} = 5.5 V V_{IN} = Gnd	All		10	µA
High level input leakage current (all input pins)	I_{IH}	V_{CC} = 5.5 V V_{IN} = 5.5 V	All		10	µA
Output leakage current	I_{LO}	$\overline{CS}$ = V_{IH}, V_{CC} = 5.5 V V_{OUT} = Gnd to 4.5 V	All		50	µA
Power supply current	I_{CC}	V_{CC} = 5.5 V $\overline{CS}$ = V_{IL}, outputs open	01		160	mA
			02		100	mA
			03		180	mA
			04		70	mA
			05,06,07		180	mA
Standby current	I_{SB}	V_{CC} = 4.5 V to 5.5 V $\overline{CS}$ = V_{IH}	01,03, 05,07		30	mA
Output low voltage	V_{OL}	I_{OL} = 2.1 mA	02,04		0.4	V
		I_{OL} = 8.0 mA	01,03, 05,06,07		0.4	V
Output high voltage	V_{OH}	I_{OH} = -1.0 mA	02,04	2.4		V
		I_{OH} = -4.0 mA	01,03, 05,06,07	2.4		V
Output short circuit current 3/ 4/	I_{OS}	V_{CC} = 5.5 V, V_{OUT} = Gnd	All		300	mA
Input capacitance 4/	C_{IN}	V_{IN} = 0 V, f = 1 MHz T_C = 25°C	All		5	pF
Peak power on 4/	I_{PO}	V_{CC} = 4.5 V, $\overline{CS}$ = 2.4 V	01,03, 05,07		70	mA
			06		50	mA
Read cycle time 2	t_{RC}	See table III	01	85		ns
			02	450		ns
			03	70		ns
			04	250		ns
			05	55		ns
			06	70		ns
			07	45		ns
Address access time	t_{AA}	See table III	01		85	ns
			02		450	ns
			03		70	ns
			04		250	ns
			05		55	ns
			06		70	ns
			07		45	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ 2/ 5/	Device type	Limits Min	Limits Max	Unit
Chip select access time 5/	t_{ACS1}	See table III	01		85	ns
			02		120	ns
			03		70	ns
			04		85	ns
			05		55	ns
			06		70	ns
			07		45	ns
Chip select access time 6/	t_{ACS2}		01		100	ns
			03		80	ns
			05		65	ns
			06		80	ns
			07		55	ns
Chip select to output in Low-Z 4/ 7/	t_{LZ}		01,03, 04,05,07	10		ns
			02,06	20		ns
Chip deselect to output in High-Z 4/ 7/	t_{HZ}		01,03	0	40	ns
			02	0	100	ns
			04	0	60	ns
			05,07	0	30	ns
			06	0	20	ns
Output hold from address change 4/	t_{OH}		01,03, 05,06,07	5		ns
			02	50		ns
			04	15		ns
Chip select to power up time 4/	t_{PU}		01,03, 05,06,07	0		ns
Chip deselect to power down time 4/	t_{PD}		01,06		30	ns
			03,05,07		20	ns
Write cycle time	t_{WC}		01	85		ns
			02	450		ns
			03	70		ns
			04	250		ns
			05	55		ns
			06	70		ns
			07	45		ns
Chip select to end of write	t_{CW}		01	70		ns
			03	55		ns
			05,07	45		ns
			06	65		ns
Address valid to end of write	t_{AW}		01	70		ns
			03	55		ns
			05,07	45		ns
			06	65		ns
Address setup time	t_{AS}		01,03, 05,06,07	0		ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ 2/ 5/	Device type	Limits Min	Limits Max	Unit
Write pulse width	t_{WP}	See table III	01	55		ns
			02	200		ns
			03	40		ns
			04	135		ns
			05	25		ns
			06	50		ns
			07	30		ns
Write recovery time	t_{WR}		01,03	15		ns
			02,04	0		ns
			05,07	10		ns
			06	5		ns
Data valid to end of write	t_{DW}		01	35		ns
			02	200		ns
			03	30		ns
			04	135		ns
			05,06	25		ns
			07	30		ns
Data hold time	t_{DH}		01,03, 05,07	10		ns
			02,04,06	0		ns
Write enabled to output in High-Z 4/ 7/	t_{WZ}		01	0	65	ns
			02	0	100	ns
			03	0	35	ns
			04	0	60	ns
			05,06,07	0	25	ns
Output active from end of write 4/ 7/	t_{OW}		01,03, 05,06,07	0		ns

1/ Output levels are tested in static state and are specified over voltage range of V_{CC}.

2/ Unless otherwise specified, the dynamic load shall be in accordance with figure 6 (load A).

3/ Duration not to exceed 1 second.

4/ Not tested.

5/ Complete terminal conditions are as specified in table III.

6/ Chip deselected for a finite time that is less than 55 ns prior to selection. (If the deselect time is 0 ns, the chip is by definition selected and access occurs according to read cycle no. 1.)

7/ Transition is measured ±500 mV from steady state voltage using figure 6 (load B).

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	2,8*	2,8*	None
Final electrical test parameters (method 5004)	1**,2,3, 7**,8	1**,2,3, 7**,8	1,7
Group A test requirements (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7,8	1,2,7
Group B end-point electrical parameters (method 5005)	1,2,3,7,8, 9,10,11	N/A	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,7	1,7
Additional electrical subgroups for group C (method 5005)	N/A	9,10,11	9,10,11
Group D end-point electrical parameters (method 5005)	1,2,3,7,8	1,7	1,7

* Maximum temperature only.

** PDA applies to subgroups 1 and 7 (see 4.2d).

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows: A special subgroup shall be added using an LTPD of 15 for class S and B devices. This subgroup shall consist of a high-voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady state life test (method 1005 of MIL-STD-883) conditions, or equivalent:

1. Test condition C, D or E, using the circuit shown on figure 4 or 5, or equivalent.

2. T_A = 125°C minimum.

3. Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Subgroups 3, 4, and 5 shall be added to the group C inspection requirements for class B and C devices, and shall consist of the conditions and limits as specified for subgroups 9, 10, and 11 of group A. These additional tests shall consist of performing the specified tests using the GALPAT pattern and then repeating using the GALRESH pattern as specified in the appendix.

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

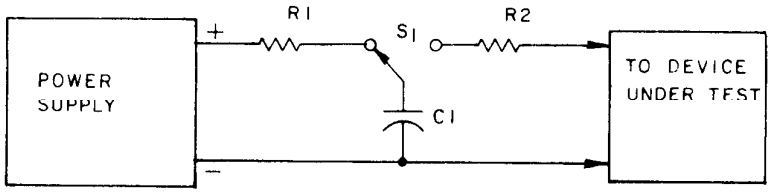
4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the steady state life or burn-in test condition, they shall be cooled to 35°C or less prior to removal of the bias.

4.5.3 High voltage (V_{ZAP}) test of input protection circuits. One input terminal of the device under test (DUT) shall be subjected to a voltage pulse of 170 volts from a 100 pF source in the following test sequence. This is a destructive test.

a. Measure I_{IH} and I_{IL} at one input terminal of the DUT at 25°C. These measurements shall be made in accordance with table III.

b. In the circuit below, charge the capacitor to +170 volts. Then, using the same terminal of the device as selected above for leakage measurements, switch the capacitor to discharge into the device terminal. Then repeat the procedure with -170 volts.

c. Within 24 hours, repeat the I_{IH} and I_{IL} measurements on the same terminal as performed above. At this time a DUT exhibiting leakage currents in excess of the specified limits is defective.



R1 = Appropriate current limiting resistance.
R2 = 1.5 kΩ ±5%.
C1 = 100 pF ±20%.
Power supply voltage = V_{ZAP} = +170 Vdc and -170 Vdc

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510. The devices covered by this specification require electrostatic protection.

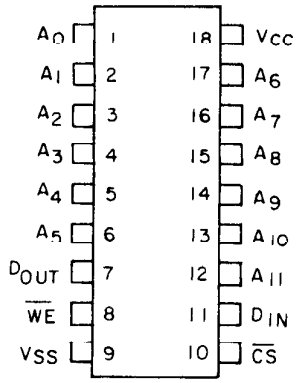
6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

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Device types 01, 03, 05 and 07

Case V



Device types 02, 04 and 06

Case V

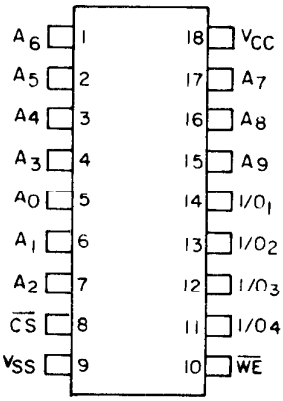
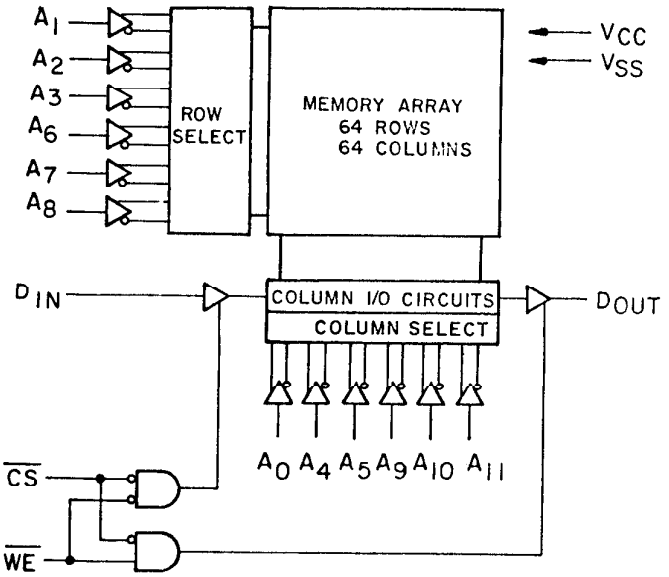


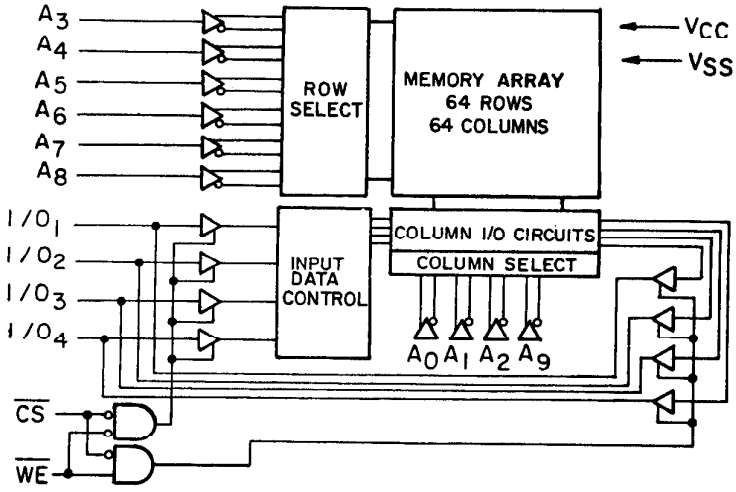
FIGURE 1. Terminal connections.

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Device types 01, 03, 05 and 07



Device types 02, 04 and 06



NOTE: Address numbering may vary between vendors.

FIGURE 2. Block diagrams.

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Device types 01, 02, 03, 04, 05, 06 and 07

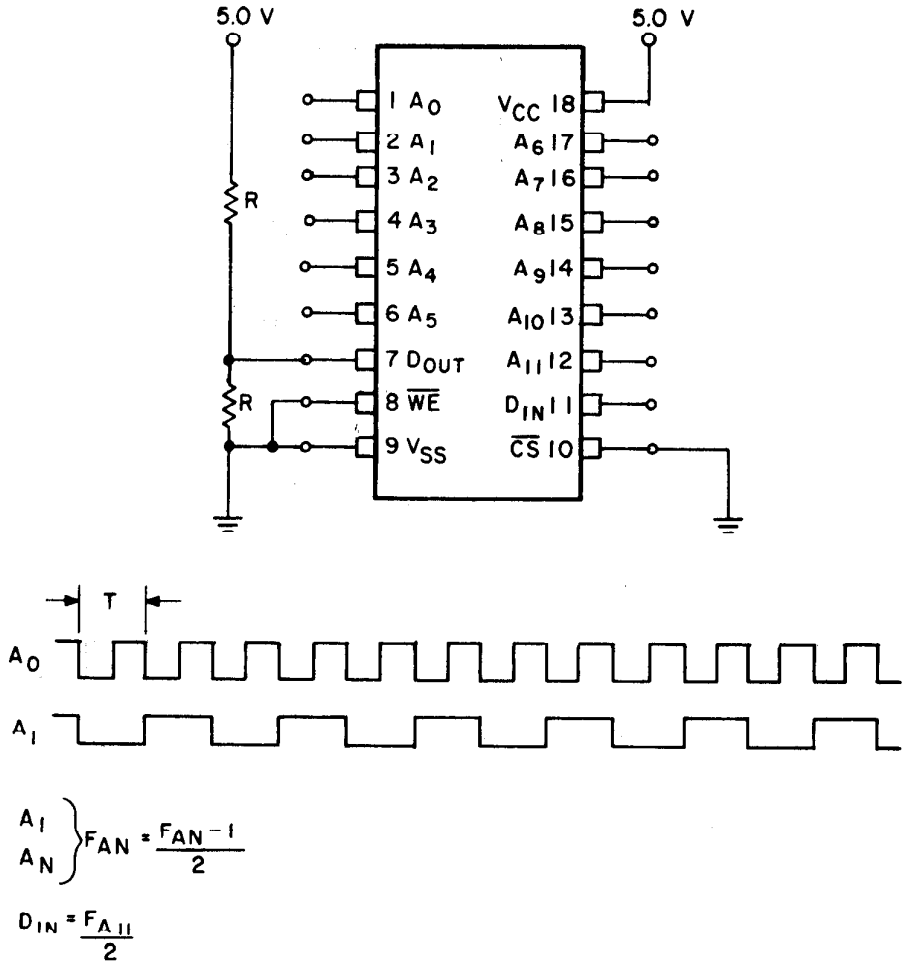
$\overline{CS}$	$\overline{WE}$	Mode	01,03,05,07 Output	01,03,05,06,07 Power	02,04,06 I/O
H	X	Not selected	High Z	Stand by	High Z
L	L	Write	High Z	Active	D_{IN}
L	H	Read	D_{OUT}	Active	D_{OUT}

H = High voltage level.
L = Low voltage level.
X = Don't care (high or low).

FIGURE 3. Truth table.

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Device types 01, 03, 05 and 07

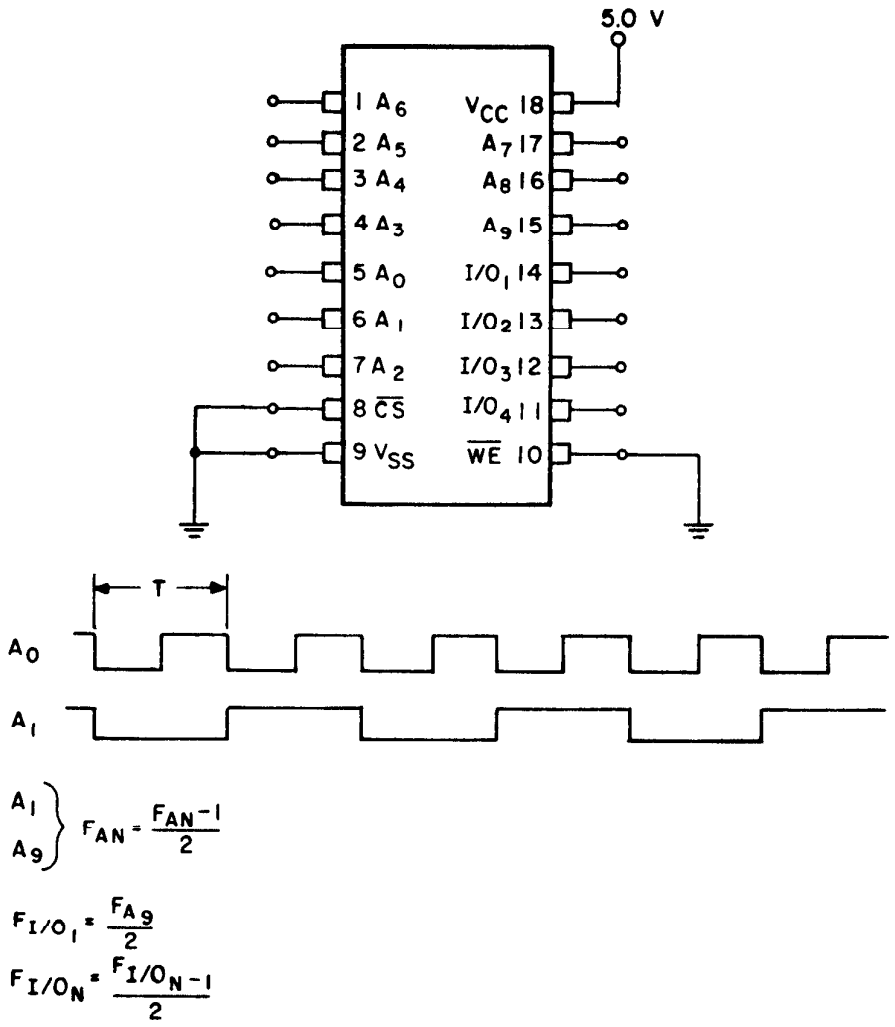


NOTES:
1. T = 20 µs, 50% duty cycle, $t_{TLH} = t_{THL} \leq 150$ ns.
2. R = 200Ω ±5%.
3. Address to be cycled in a binary sequence where A_0 is least significant and A_{11} is most significant.

FIGURE 4. Burn-in and steady state life test circuit and waveforms.

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Device types 02, 04 and 06

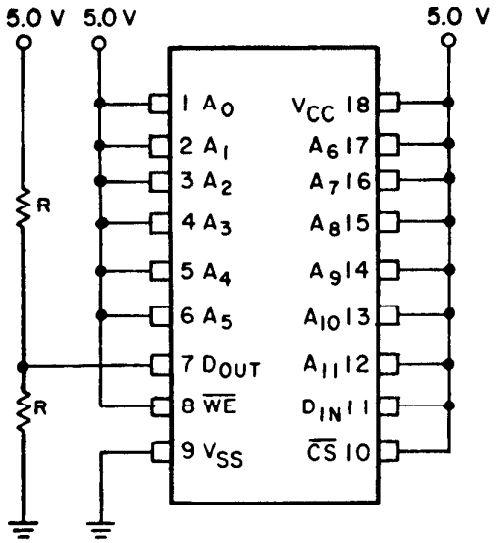


NOTES:
1. T = 20 µs, 50% duty cycle, $t_{TLH} = t_{THL} \leq 150$ ns.
2. Address to be cycled in a binary sequence where A_0 is least significant and A_9 is most significant.

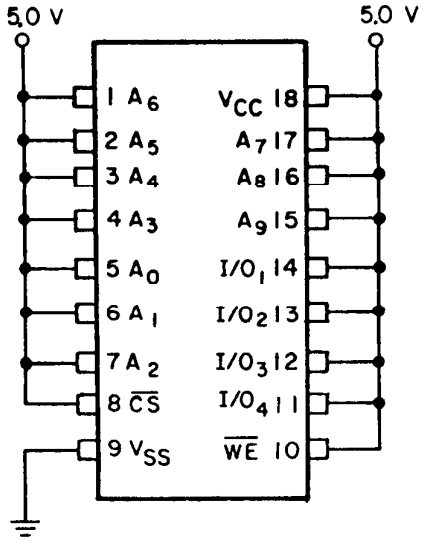
FIGURE 4. Burn-in and steady state life test circuit and waveforms - Continued.

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Check the source to verify that this is the current version before use.

Device types 01, 03, 05 and 07



Device types 02, 04 and 06

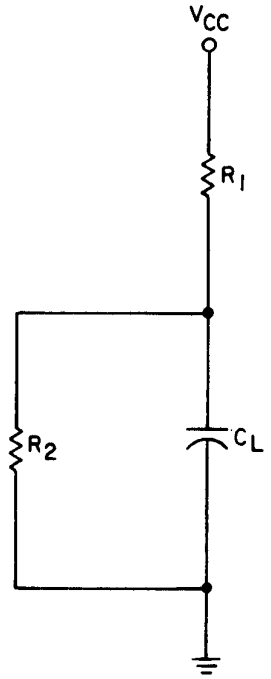


NOTE: R = 1 kΩ ±5%.

FIGURE 5. Optional steady state life test circuit.

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Device types 01, 03, 05, 06 and 07

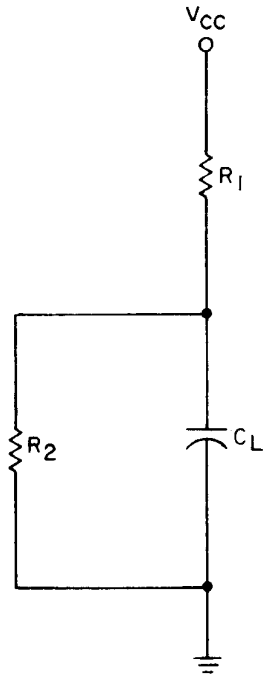


NOTES:
1. V_{CC} is defined in table III.
2. Load A: R1 = 480Ω ±5%; R2 = 255Ω ±5%; and C_L = 30 pF (including probe and jig capacitance).
3. Load B: R1 = 480Ω ±5%, R2 = 255Ω ±5%; and C_L = 5 pF (including probe and jig capacitance).

FIGURE 6. Dynamic load for switching tests.

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Device types 02 and 04

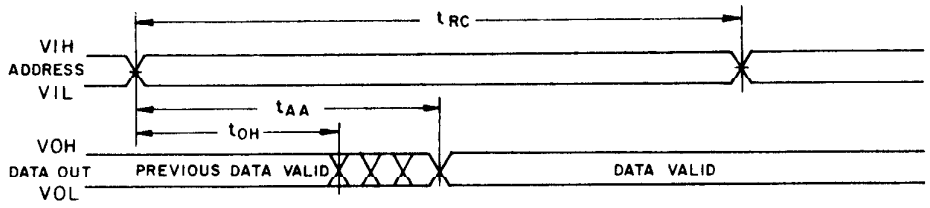


NOTES:
1. V_{CC} is defined in table III.
2. C_L = 100 pF total, including probe and jig capacitance.
3. R1 = 1.8 kΩ ±5%, R2 = 1 kΩ ±5%.

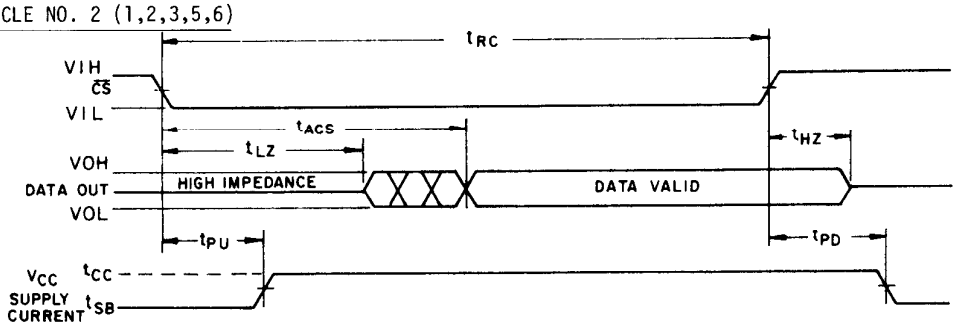
FIGURE 6. Dynamic load for switching tests - Continued.

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READ CYCLE NO. 1 (3,4,6)



READ CYCLE NO. 2 (1,2,3,5,6)



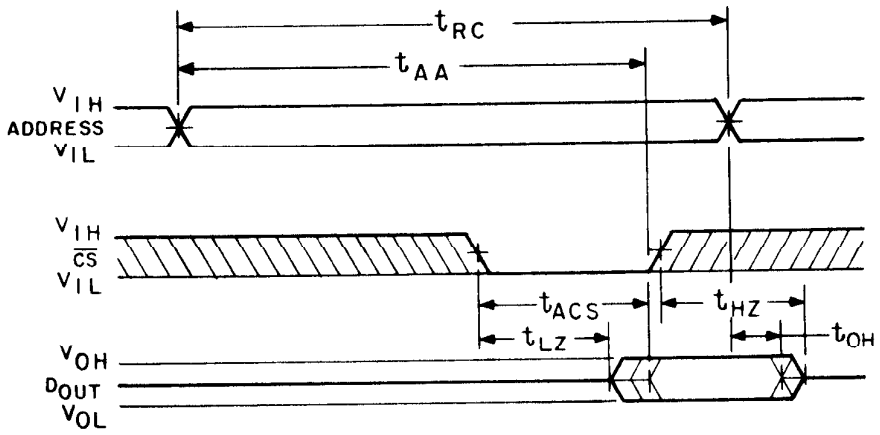
Waveforms

NOTES:
1. Chip deselected for greater than 55 ns prior to selection.
2. Chip deselected for a finite time that is less than 55 ns prior to selection. (If the deselect time is 0 ns the chip is by definition selected and access occurs according to read cycle no. 1.)
3. $\overline{WE}$ is high for read cycles.
4. Device is continuously selected, $\overline{CS} = V_{IL}$.
5. Addresses valid prior to or coincident with $\overline{CS}$ transition low.
6. See table I for limits and complete terminal conditions.
7. Input and output timing reference levels are 1.5 volts with input pulse levels of ground to 3.0 volts.
8. t_{LZ} and t_{HZ} are measured at ±500 mV from steady state with 5 pF load.

FIGURE 7. Read cycle waveforms and test conditions for device types 01, 03, 05, 06 and 07.

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READ CYCLE



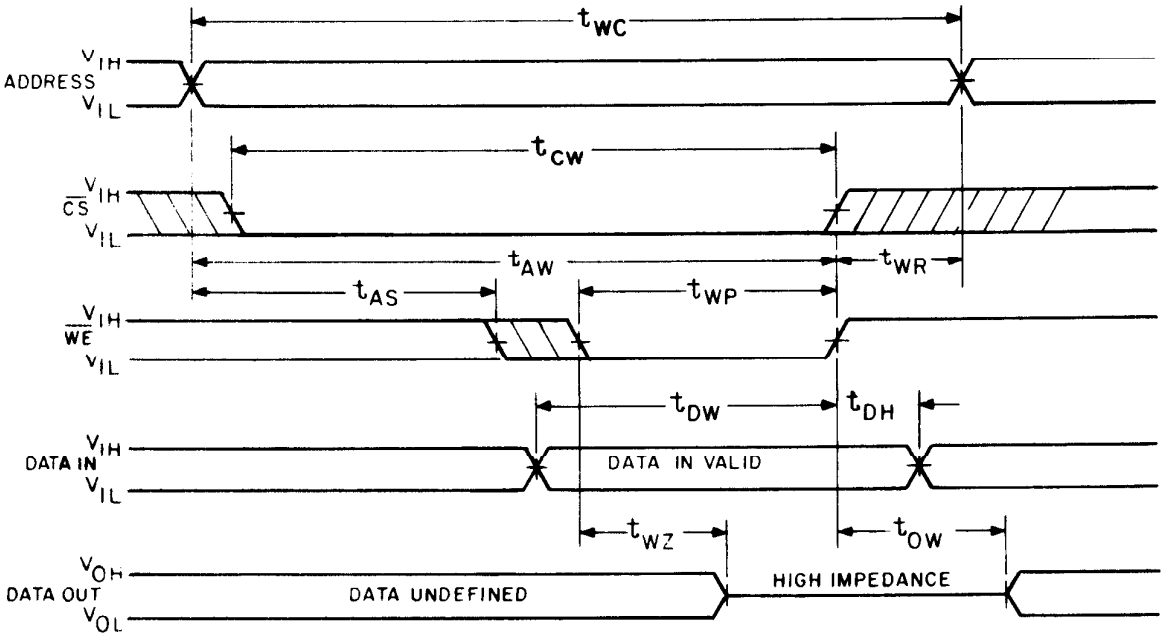
Waveforms

NOTES:
1. $\overline{WE}$ is high for a read cycle.
2. If the $\overline{CS}$ low transition occurs simultaneously with the $\overline{WE}$ low transition, the output buffers remain in a high impedance state.
3. $\overline{WE}$ must be high during all address transitions.
4. Rise and fall times of input pulses $\leq$ 10 ns.
5. Input and output timing reference levels are 1.5 volts with input pulse levels of ground to 3.0 volts.
6. t_{HZ} and t_{LZ} are measured at ±500 mV from steady state with 5 pF load.

FIGURE 7. Read cycle waveforms and test conditions for device types 02 and 04 - Continued.

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WRITE CYCLE



Waveforms

NOTES:
1. See table I for limits and complete terminal conditions.
2. Input and output timing reference levels are 1.5 volts with input pulse levels of ground to 3.0 volts.
3. t_{OW} and t_{WZ} are measured at ±500 mV from steady state with 6 pF load.

FIGURE 8. Write cycle waveforms and test conditions for device types 01, 02, 03, 04, 05, 06 and 07.

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TABLE III. Group A inspection for device types 01, 03, 05 and 07.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V test no.	1 A_0	2 A_1	3 A_2	4 A_3	5 A_4	6 A_5	7 D_{OUT}	8 $\overline{WE}$	9 V_{SS}	10 $\overline{CS}$	11 D_{IN}	12 A_{11}	13 A_{10}	14 A_9	15 A_8	16 A_7	17 A_6	18 V_{CC}	Algorithms 1/	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = 25°C	V_{OH}	3006	1	GND	GND	GND	GND	GND	GND	-4 mA	4.5 V	GND	GND	GND	GND	GND	GND	GND	GND	GND	4.5 V		D_{OUT}	2.4		V
	V_{OL}	3007	2	GND	GND	GND	GND	GND	GND	+8 mA	4.5 V	GND	GND	GND	GND	GND	GND	GND	GND	GND	4.5 V		D_{OUT}		0.4	V
	I_{IH}	3010	3	5.5 V	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 V		A_0		10	uA
			4	GND	5.5 V	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			A_1			
			5	GND	GND	5.5 V	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			A_2			
			6	GND	GND	GND	5.5 V	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			A_3			
			7	GND	GND	GND	GND	5.5 V	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			A_4			
			8	GND	GND	GND	GND	GND	5.5 V		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			A_5			
			9	GND	GND	GND	GND	GND	GND		5.5 V	GND	GND	GND	GND	GND	GND	GND	GND	GND			$\overline{WE}$			
			10	GND	GND	GND	GND	GND	GND		GND	GND	5.5 V	GND	GND	GND	GND	GND	GND	GND			$\overline{CS}$			
			11	GND	GND	GND	GND	GND	GND		GND	GND	GND	5.5 V	GND	GND	GND	GND	GND	GND			D_{IN}			
			12	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND			A_{11}			
			13	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	5.5 V	GND	GND	GND	GND			A_{10}			
			14	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	5.5 V	GND	GND	GND			A_9			
			15	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	5.5 V	GND	GND			A_8			
			16	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	5.5 V	GND			A_7			
			17	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 V			A_6			
	I_{IL}	3009	18	5.5 V	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 V		A_0		10	uA
			19	GND	5.5 V	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			A_1			
			20	GND	GND	5.5 V	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			A_2			
			21	GND	GND	GND	5.5 V	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			A_3			
			22	GND	GND	GND	GND	5.5 V	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			A_4			
			23	GND	GND	GND	GND	GND	5.5 V		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			A_5			
			24	GND	GND	GND	GND	GND	GND		5.5 V	GND	GND	GND	GND	GND	GND	GND	GND	GND			$\overline{WE}$			
			25	GND	GND	GND	GND	GND	GND		GND	GND	5.5 V	GND	GND	GND	GND	GND	GND	GND			$\overline{CS}$			
			26	GND	GND	GND	GND	GND	GND		GND	GND	GND	5.5 V	GND	GND	GND	GND	GND	GND			D_{IN}			
			27	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND			A_{11}			
			28	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	5.5 V	GND	GND	GND	GND			A_{10}			
			29	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	5.5 V	GND	GND	GND			A_9			

See footnotes at end of device types 01, 03, 05 and 07.

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TABLE III. Group A inspection for device types 01, 03, 05 and 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case temp. Test no.	1 A_0	2 A_1	3 A_2	4 A_3	5 A_4	6 A_5	7 D_{OUT}	8 $\overline{WE}$	9 V_{SS}	10 $\overline{CS}$	11 D_{IN}	12 A_{11}	13 A_{10}	14 A_9	15 A_8	16 A_7	17 A_6	18 V_{CC}	Algorithms 1/	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = 25°C	I_{I1}	3009	30	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 V		A_8		10	µA
	I_{IL}	3009	31	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 V		A_7		10	
	I_{I1}	3009	32	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 V		A_6		10	
	I_{LO1}		33							4.5 V	2.0 V		2.0 V	2.0 V							4.5 V		D_{OUT}		50	
	I_{LO1}		34							5.5 V											5.5 V		D_{OUT}			
	I_{LO2}		35							0.4 V											4.5 V		D_{OUT}			
	I_{LO2}		36							0.4 V											5.5 V		D_{OUT}			
	I_{SI}		37	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	GND	GND		4.5 V	GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V		V_{CC}		30	mA
	I_{SI}		38	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	GND		5.5 V	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		V_{CC}		30	
	I_{CC}		39	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	GND	GND	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		V_{CC}		2/	
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = 125°C.																									
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C.																									
BIT STRESS TEST			40	3/	3/	3/	3/	3/	3/	3/	3/	GND	3/	3/	3/	3/	3/	3/	3/	3/	3/	CKBD and $\overline{CKBD}$	D_{OUT}	V_{OL} ≤1.5	V_{OH} >1.5	V
T_C = 25°C	t_{AA}	Fig. 7, 8	41	4/	4/	4/	4/	4/	4/	4/	4/		4/	4/	4/	4/	4/	4/	4/	4/	4.5 V					
			42	4/	4/	4/	4/	4/	4/	4/	4/		4/	4/	4/	4/	4/	4/	4/	4/	5.5 V					
			43	5/	5/	5/	5/	5/	5/	5/	5/		5/	5/	5/	5/	5/	5/	5/	5/	4.5 V					
			44	5/	5/	5/	5/	5/	5/	5/	5/		5/	5/	5/	5/	5/	5/	5/	5/	5.5 V					
			45	6/	6/	6/	6/	6/	6/	6/	6/		6/	6/	6/	6/	6/	6/	6/	6/	4.5 V					
			46	6/	6/	6/	6/	6/	6/	6/	6/		6/	6/	6/	6/	6/	6/	6/	6/	5.5 V					
			47	7/	7/	7/	7/	7/	7/	7/	7/		7/	7/	7/	7/	7/	7/	7/	7/	4.5 V					
			48	7/	7/	7/	7/	7/	7/	7/	7/		7/	7/	7/	7/	7/	7/	7/	7/	5.5 V					
			49	8/	8/	8/	8/	8/	8/	8/	8/		8/	8/	8/	8/	8/	8/	8/	8/	4.5 V					
			50	8/	8/	8/	8/	8/	8/	8/	8/		8/	8/	8/	8/	8/	8/	8/	8/	5.5 V					
			51	9/	9/	9/	9/	9/	9/	9/	9/		9/	9/	9/	9/	9/	9/	9/	9/	4.5 V					
			52	9/	9/	9/	9/	9/	9/	9/	9/		9/	9/	9/	9/	9/	9/	9/	9/	5.5 V					
	t_{AS1}		53	10/	10/	10/	10/	10/	10/	10/	10/		10/	10/	10/	10/	10/	10/	10/	10/	5.5 V					
	t_{AS2}		54	10/	10/	10/	10/	10/	10/	10/	10/		10/	10/	10/	10/	10/	10/	10/	10/	5.5 V					

See footnotes at end of device type 01, 03, 05 and 07.

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TABLE III. Group A inspection for device types 01, 03, 05 and 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.3 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V Test no.	1 A_0	2 A_1	3 A_2	4 A_3	5 A_4	6 A_5	7 D_{OUT}	8 $\overline{WE}$	9 V_{SS}	10 $\overline{CS}$	11 D_{IN}	12 A_{11}	13 A_{10}	14 A_9	15 A_8	16 A_7	17 A_6	18 V_{CC}	Algorithms 1/	Measured terminal	Test limits Min	Test limits Max	Unit
7 T_C = 25°C	t_{ACS1}	Fig. 7, 8	55	10/	10/	10/	10/	10/	10/	10/	10/	GND	10/	10/	10/	10/	10/	10/	10/	10/	4.5 V	CKBD and $\overline{CKBD}$	D_{OUT}	V_{OL} <1.5	V_{OH} >1.5	V
7 T_C = 25°C	t_{ACS2}	Fig. 7, 8	56	10/	10/	10/	10/	10/	10/	10/	10/	GND	10/	10/	10/	10/	10/	10/	10/	10/	4.5 V	CKBD and CKBD	D_{OUT}	V_{OL} <1.5	V_{OH} >1.5	V
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = 125°C and -55°C.																									
9 T_C = 25°C	t_{AA}	Fig. 7, 8	57	11/	11/	11/	11/	11/	11/	11/	11/	GND	11/	11/	11/	11/	11/	11/	11/	11/	4.5 V	GALPAT and GALRESH	D_{OUT}		12/	ns
	t_{AA}		58																		5.5 V				12/	
	t_{ACS1}		59																		4.5 V				13/	
	t_{ACS1}		60																		5.5 V				13/	
	t_{ACS2}		61																		4.5 V				14/	
	t_{ACS2}		62																		5.5 V				14/	
	t_{WC}		63																		4.5 V			15/		
	t_{WC}		64																		5.5 V			15/		
	t_{CW}		65																		4.5 V			16/		
	t_{CW}		66																		5.5 V			16/		
	t_{AW}		67																		4.5 V			17/		
	t_{AW}		68																		5.5 V			17/		
	t_{AS}		69																		4.5 V			0		
	t_{AS}		70																		5.5 V			0		
	t_{WP}		71																		4.5 V			18/		
	t_{WP}		72																		5.5 V			18/		
	t_{WR}		73																		4.5 V			19/		
	t_{WR}		74																		5.5 V			19/		
	t_{DW}		75																		4.5 V			20/		
	t_{DW}		76																		5.5 V			20/		
	t_{DH}		77																		4.5 V			10		
	t_{DH}		78																		5.5 V			10		

See footnotes at end of device types 01, 03, 05 and 07.

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TABLE II. Group A inspection for device types 01, 03, 05 and 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Algorithm 1/	Measured terminal	Test limits		Unit
			Test no.	A_0	A_1	A_2	A_3	A_4	A_5	D_{OUT}	$\overline{WE}$	V_{SS}	$\overline{CS}$	D_{IN}	A_{11}	A_{10}	A_9	A_8	A_7	A_6	V_{CC}			Min	Max	
10	Same tests, terminal conditions, and limits as subgroup 9, except T_C = 125°C.																									
11	Same tests, terminal conditions, and limits as subgroup 9, except T_C = -55°C.																									

1/ See appendix for description of algorithms.

2/ I_{CC} = 160 for device type 01; 180 for device types 03, 05 and 07.

3/ V_{IL} = GND, V_{IH} = 6.0 V, pause time = 250 ms/loop max, $\overline{CS}$ = high, only performed once at 125°C, and V_{CC} = 0.7 V min.

4/ V_{IL} = 0.8 V and V_{IH} = 2.0 V.

5/ Algorithm has 60 ms where chip is deselected between the write.

6/ V_{IL} = GND, V_{IH} = 3.0 V, and all address setup times are at minimums.

7/ V_{IL} = GND, V_{IH} = 3.0 V, and all write pulse timing, are at minimums.

8/ V_{IL} = GND, V_{IH} = 3.0 V, and all address ending timing are at minimums.

9/ V_{IL} = GND, V_{IH} = 3.0 V, and t_{AA} is measured at minimum timing.

10/ V_{IL} = GND, V_{IH} = 3.0 V, t_{ACS1} and t_{ACS2} are measured at minimum timing.

11/ V_{IL} = 0.8 V, V_{IH} = 2.0 V, and all parameters are measured at minimum timing.

12/ t_{AA} = 85 ns for device type 01; 70 ns for device type 03; 55 ns for device type 05; 45 ns for device type 07.

13/ t_{ACS1} = 85 ns for device type 01; 70 ns for device type 03; 55 ns for device type 05; 45 ns for device type 07.

14/ t_{ACS2} = 100 ns for device type 01; 80 ns for device type 03; 65 ns for device type 05; 55 ns for device type 07.

15/ t_{WC} = 85 ns for device type 01; 70 ns for device type 03; 55 ns for device type 05; 45 ns for device type 07.

16/ t_{CW} = 70 ns for device type 01; 55 ns for device type 03; 45 ns for device types 05 and 07.

17/ t_{AW} = 70 ns for device type 01; 55 ns for device type 03; 45 ns for device types 05 and 07.

18/ t_{WP} = 55 ns for device type 01; 40 ns for device type 03; 25 ns for device type 05; 30 ns for device type 07.

19/ t_{WR} = 15 ns for device types 01 and 03; 10 ns for device types 05 and 07.

20/ t_{DW} = 35 ns for device type 01; 30 ns for device type 03; 25 ns for device type 05; 30 ns for device type 07.

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TABLE III. Group A inspection for device types 02 and 04.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V Test no.	1 A_0	2 A_5	3 A_4	4 A_3	5 A_2	6 A_1	7 A_2	8 $\overline{CS}$	9 V_{SS}	10 $\overline{WE}$	11 I/O_4	12 I/O_3	13 I/O_2	14 I/O_1	15 A_9	16 A_1	17 A_7	18 V_{CC}	Algorithms 1/	Measured terminal	Test limits Min	Test limits Max	Unit
1/ $T_C = T_A = 25°C$	V_{OH}	3006	1	GND	GND	GND	GND	GND	GND	GND	GND	GND	3.0 V	-1.0 mA				GND	GND	GND	4.5 V		I/O_4	2.4		V
			2												-1.0 mA								I/O_3			
			3													-1.0 mA							I/O_2			
			4														-1.0 mA						I/O_1			
	V_{OL}	3007	5											2.1 mA									I/O_4		0.4	
			6												2.1 mA								I/O_3			
			7													2.1 mA							I/O_2			
			8														2.1 mA						I/O_1			
	I_{I1}	3010	9	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	5.5 V	GND	GND	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V		A_6		10	µA
			10	GND	GND	GND	GND	GND	GND	GND	GND		GND	5.5 V	5.5 V	5.5 V	5.5 V	GND	GND	GND	GND		A_5			
			11	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND		5.5 V	GND	GND	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V		A_4			
			12	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V		GND	5.5 V	5.5 V	5.5 V	5.5 V	GND	GND	GND	GND		A_3			
			13	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND		5.5 V	GND	GND	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V		A_0			
			14	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V		GND	5.5 V	5.5 V	5.5 V	5.5 V	GND	GND	GND	GND		A_1			
			15	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND		5.5 V	GND	GND	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V		A_2			
			16	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V		GND	5.5 V	5.5 V	5.5 V	5.5 V	GND	GND	GND	GND		$\overline{CS}$			
			17	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND		5.5 V	GND	GND	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V		$\overline{WE}$			
			18	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V		GND	5.5 V	5.5 V	5.5 V	5.5 V	GND	GND	GND	GND		I/O_4			
			19	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND		5.5 V	GND	GND	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V		I/O_3			
			20	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V		GND	5.5 V	5.5 V	5.5 V	5.5 V	GND	GND	GND	GND		I/O_2			
			21	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND		5.5 V	GND	GND	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V		I/O_1			
			22	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V		GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	5.5 V	GND		A_9			
			23	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND		5.5 V	GND	GND	GND	GND	GND	5.5 V	GND	5.5 V		A_8			
			24	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V		GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	5.5 V	GND		A_7			
	I_{IL}	3009	25	GND	GND	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	5.5 V		A_6		10	µA
			26																				A_5			
			27																				A_4			
			28																				A_3			
			29																				A_0			

See footnotes at end of device types 02 and 04.

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TABLE III. Group A inspection for device types 02 and 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Algorithm 1/	Measured terminal	Test limits Min	Test limits Max	Unit
				A_6	A_5	A_4	A_3	A_0	A_1	A_2	$\overline{CS}$	V_{SS}	$\overline{WE}$	I/O_4	I/O_3	I/O_2	I/O_1	A_9	A_8	A_7	V_{CC}					
T_C = 25°C 1/	I_{IL}	3009	30	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 V		A_1		10	µA
			31																				A_2			
			32																				$\overline{CS}$			
			33																				$\overline{WE}$			
			34																				I/O_4			
			35																				I/O_3			
			36																				I/O_2			
			37																				I/O_1			
			38																				A_9			
			39																				A_8			
			40																				A_7			
	I_{LO1}		41								3.0 V		3.0 V										I/O_4		50	
			42												GND								I/O_3			
			43													GND							I/O_2			
			44														GND						I/O_1			
	I_{LO2}		45											4.5 V									I/O_4			
			46												4.5 V								I/O_3			
			47													4.5 V							I/O_2			
			48														4.5 V						I/O_1			
	I_{CC}	3005	49											GND	GND	GND	GND	GND	GND	GND			V_{CC}		2/	mA
HIT STRESS TEST			50	3/	3/	3/	3/	3/	3/	3/	3/		3/	3/	3/	3/	3/	3/	3/	3/	3/		CKBD and $\overline{CKBD}$			V
T_C = 25°C 2/	t_{AA}	Fig. 7, 8	51	4/	4/	4/	4/	4/	4/	4/	4/		4/	4/	4/	4/	4/	4/	4/	4/	4/		All I/O	V_{IL} ≤1.5	V_{OH} ≥1.5	
			52	4/	4/	4/	4/	4/	4/	4/	4/		5/	4/	4/	4/	4/	4/	4/	4/	5/					
			53	5/	5/	5/	5/	5/	5/	5/	5/		5/	5/	5/	5/	5/	5/	5/	5/	5/					
			54	5/	5/	5/	5/	5/	5/	5/	5/		5/	5/	5/	5/	5/	5/	5/	5/	5/					
			55	6/	6/	6/	6/	6/	6/	6/	6/		6/	6/	6/	6/	6/	6/	6/	6/	6/					
			56	6/	6/	6/	6/	6/	6/	6/	6/		6/	6/	6/	6/	6/	6/	6/	6/	6/					

See footnotes at end of device types 02 and 04.

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TABLE III. Group A inspection for device types 02 and 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.2 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V Test no.	1 A_6	2 A_5	3 A_4	4 A_3	5 A_0	6 A_1	7 A_2	8 $\overline{CS}$	9 V_{SS}	10 $\overline{WE}$	11 I/O_4	12 I/O_3	13 I/O_2	14 I/O_1	15 A_9	16 A_8	17 A_7	18 V_{CC}	Algorithms 1/	Measured terminal	Test limits Min	Test limits Max	Unit
7 T_C = 25°C	t_{AA}	Fig. 7, 8	57	7/	7/	7/	7/	7/	7/	7/	7/	GND	7/	7/	7/	7/	7/	7/	7/	7/	4.5 V	CKBD and $\overline{CKBD}$	All I/O	V_{OL} <1.5	V_{OH} >1.5	V
			58	7/	7/	7/	7/	7/	7/	7/	7/		7/	7/	7/	7/	7/	7/	7/	7/	5.5 V					
			59	8/	8/	8/	8/	8/	8/	8/	8/		8/	8/	8/	8/	8/	8/	8/	8/	4.5 V					
			60	8/	8/	8/	8/	8/	8/	8/	8/		8/	8/	8/	8/	8/	8/	8/	8/	5.5 V					
			61	9/	9/	9/	9/	9/	9/	9/	9/		9/	9/	9/	9/	9/	9/	9/	9/	4.5 V					
			62	9/	9/	9/	9/	9/	9/	9/	9/		9/	9/	9/	9/	9/	9/	9/	9/	5.5 V					
	t_{ACS}		63	10/	10/	10/	10/	10/	10/	10/	10/		10/	10/	10/	10/	10/	10/	10/	10/	4.5 V					
	t_{ACS}		64	10/	10/	10/	10/	10/	10/	10/	10/		10/	10/	10/	10/	10/	10/	10/	10/	5.5 V					
	t_{ACS}		65	10/	10/	10/	10/	10/	10/	10/	10/		10/	10/	10/	10/	10/	10/	10/	10/	4.5 V	MARCH				
	t_{ACS}		66	10/	10/	10/	10/	10/	10/	10/	10/		10/	10/	10/	10/	10/	10/	10/	10/	5.5 V	MARCH				
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = 125°C and T_C = -55°C.																								
9 T_C = 25°C	t_{AA}	Fig. 7, 8	67	11/	11/	11/	11/	11/	11/	11/	11/	GND	11/	11/	11/	11/	11/	11/	11	11/	4.5 V	GALPAT and GALRESH	All I/O		12/	ns
	t_{AA}		68																		5.5 V				12/	
	t_{ACS1}		69																		4.5 V				13/	
	t_{ACS1}		70																		5.5 V				13/	
	t_{WC}		71																		4.5 V			14/		
	t_{WC}		72																		5.5 V			14/		
	t_{WP}		73																		4.5 V			15/		
	t_{WP}		74																		5.5 V			15/		
	t_{WR}		75																		4.5 V			0		
	t_{WR}		76																		5.5 V			0		
	t_{DW}		77																		4.5 V			16/		
	t_{DW}		78																		5.5 V			16/		

See footnotes at end of device types 02 and 04.

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TABLE III. Group A inspection for device types 02 and 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Algorithms 1/	Measured terminal	Test limits Min	Test limits Max	Unit
				A_6	A_5	A_4	A_3	A_0	A_1	A_2	$\overline{CS}$	V_{SS}	$\overline{WE}$	I/O_4	I/O_3	I/O_2	I/O_1	A_9	A_8	A_7	V_{CC}					
9 T_C = 25°C	t_{DH}	Fig. 7, 8	79	11/	11/	11/	11/	11/	11/	11/	11/	GND	11/	11/	11/	11/	11/	11/	11/	11/	4.5 V	GALPAT and GALRESH	All I/O	0		ns
9 T_C = 25°C	t_{DH}	Fig. 7, 8	80	11/	11/	11/	11/	11/	11/	11/	11/	GND	11/	11/	11/	11/	11/	11/	11/	11/	4.5 V	GALPAT and GALRESH	All I/O	0		ns
10	Same tests, terminal conditions, and limits as subgroup 9, except T_C = 125°C.																									
11	Same tests, terminal conditions, and limits as subgroup 9, except T_C = -55°C.																									

1/ See appendix for description of algorithms.

2/ I_{CC} = 100 mA for device type 02; 90 mA for device type 04.

3/ V_{IL} = GND, V_{IH} = 3.0 V, cycle time = 250 ns/loop max, $\overline{CS}$ = high, only performed once at 125°C, and V_{CC} = 7.0 V min.

4/ V_{IL} = 0.8 V and V_{IH} = 2.0 V.

5/ Algorithm has 60 ns where chip is deselected between the write.

6/ V_{IL} = GND, V_{IH} = 3.0 V, and all address setup times are at minimums.

7/ V_{IL} = GND, V_{IH} = 3.0 V, and all write pulse timing are at minimums.

8/ V_{IL} = GND, V_{IH} = 3.0 V, and all address ending timing are at minimums.

9/ V_{IL} = GND, V_{IH} = 3.0 V, and t_{AA} is measured at minimum timing.

10/ V_{IL} = GND, V_{IH} = 3.0 V, t_{ACS1} and t_{ACS2} are measured at minimum timing.

11/ V_{IL} = 0.8 V, V_{IH} = 2.0 V, and all parameters are measured at minimum timing.

12/ t_{AA} = 450 ns for device type 02; 250 ns for device type 04.

13/ t_{ACS1} = 120 ns for device type 02; 85 ns for device type 04.

14/ t_{WC} = 450 ns for device type 02; 250 ns for device type 04.

15/ t_{WP} = 200 ns for device type 02; 135 ns for device type 04.

16/ t_{DW} = 200 ns for device type 02; 135 ns for device type 04.

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TABLE III. Group A inspection for device type 06.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V Test no.	1 A_6	2 A_5	3 A_4	4 A_3	5 A_2	6 A_1	7 A_0	8 $\overline{CS}$	9 V_{SS}	10 $\overline{WE}$	11 I/O_4	12 I/O_3	13 I/O_2	14 I/O_1	15 A_9	15 A_8	17 A_7	18 V_{CC}	Algorithms 1/	Measured terminal	Test limits Min	Test limits Max	Unit
T_C = T_A = 25°C	V_{OH}	3006	1	GND	GND	GND	GND	GND	GND	GND	GND	GND	3.0 V	−4 mA	−4 mA	−4 mA	−4 mA	GND	GND	GND	4.5 V		I/O_4	2.4		V
			2																				I/O_3			
			3																				I/O_2			
			4																				I/O_1			
	V_{OL}	3007	5											8 mA	8 mA	8 mA	8 mA						I/O_4		0.4	
			6																				I/O_3			
			7																				I/O_2			
			8																				I/O_1			
	I_{IH}	3010	9	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	5.5 V	GND	GND	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V		A_6		10	μA
			10	GND	GND	GND	GND	GND	GND	GND	GND		GND										A_5			
			11																				A_4			
			12																				A_3			
			13																				A_2			
			14																				A_1			
			15																				A_0			
			16																				$\overline{CS}$			
			17																				$\overline{WE}$			
			18																				I/O_4			
			19																				I/O_3			
			20																				I/O_2			
			21																				I/O_1			
			22																				A_9			
			23																				A_8			
			24																				A_7			
	I_{IL}	3009	25																				A_6			
			26																				A_5			
			27																				A_4			
			28																				A_3			
			29																				A_0			

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Algorithms 1/	Measured terminal	Test limits Min	Test limits Max	Unit
				A_6	A_5	A_4	A_3	A_0	A_1	A_2	$\overline{CS}$	V_{SS}	$\overline{WE}$	I/O_4	I/O_3	I/O_2	I/O_1	A_9	A_8	A_7	V_{CC}					
1/ T_C = 25°C	I_{IL}	3009	30	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 V		A_1		10	µA
			31																				A_2			
			32																				$\overline{CS}$			
			33																				$\overline{WE}$			
			34																				I/O_4			
			35																				I/O_3			
			36																				I/O_2			
			37																				I/O_1			
			38																				A_9			
			39																				A_8			
			40																				A_7			
	I_{LO1}		41								3.0 V		3.0 V										I/O_4		50	
			42												GND								I/O_3			
			43													GND							I/O_2			
			44														GND						I/O_1			
	I_{LO2}		45											4.5 V									I/O_4			
			46												4.5 V								I/O_3			
			47													4.5 V							I/O_2			
			48														4.5 V						I/O_1			
	I_{CC}	3005	49								GND		GND	GND	GND	GND	GND						V_{CC}		2/	mA
BIT STRESS TEST			50	3/	3/	3/	3/	3/	3/	3/	3/		3/	3/	3/	3/	3/	3/	3/	3/	3/	1/	CKBD and $\overline{CKBD}$	All I/O	V_{OL} ≤1.5	I_{OH} ≥1.5	V
7/ T_C = 25°C	t_{AA}	Fig. 7, 8	51	4/	4/	4/	4/	4/	4/	4/	4/		4/	4/	4/	4/	4/	4/	4/	4/	4/	4.5 V				
			52	4/	4/	4/	4/	4/	4/	4/	4/		4/	4/	4/	4/	4/	4/	4/	4/	4/	5.5 V				
			53	5/	5/	5/	5/	5/	5/	5/	5/		5/	5/	5/	5/	5/	5/	5/	5/	5/	4.5 V				
			54	5/	5/	5/	5/	5/	5/	5/	5/		5/	5/	5/	5/	5/	5/	5/	5/	5/	5.5 V				
			55	6/	6/	6/	6/	6/	6/	6/	6/		6/	6/	6/	6/	6/	6/	6/	6/	6/	4.5 V				
			56	6/	6/	6/	6/	6/	6/	6/	6/		6/	6/	6/	6/	6/	6/	6/	6/	6/	5.5 V				

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V test no.	1 A_6	2 A_5	3 A_4	4 A_3	5 A_0	6 A_1	7 A_2	8 $\overline{CS}$	9 V_{SS}	10 $\overline{WE}$	11 I/O_4	12 I/O_3	13 I/O_2	14 I/O_1	15 A_9	16 A_8	17 A_7	18 V_{CC}	Algorithms 1/	Measured terminal	Test limits Min	Test limits Max	Unit
7 T_C = 25°C	t_{AA}	Fig. 7, 8	57	7/	7/	7/	7/	7/	7/	7/	7/	GND	7/	7/	7/	7/	7/	7/	7/	7/	4.5 V	CKBD and $\overline{CKBD}$	All I/O	V_{OL} <1.5	V_{OH} >1.5	V
			58	7/	7/	7/	7/	7/	7/	7/	7/		7/	7/	7/	7/	7/	7/	7/	7/	5.5 V					
			59	8/	8/	8/	8/	8/	8/	8/	8/		8/	8/	8/	8/	8/	8/	8/	8/	4.5 V					
			60	8/	8/	8/	8/	8/	8/	8/	8/		8/	8/	8/	8/	8/	8/	8/	8/	5.5 V					
			61	9/	9/	9/	9/	9/	9/	9/	9/		9/	9/	9/	9/	9/	9/	9/	9/	4.5 V					
			62	9/	9/	9/	9/	9/	9/	9/	9/		9/	9/	9/	9/	9/	9/	9/	9/	5.5 V					
	t_{ACS}		63	10/	10/	10/	10/	10/	10/	10/	10/		10/	10/	10/	10/	10/	10/	10/	10/	5.5 V					
	t_{ACS1}		64	10/	10/	10/	10/	10/	10/	10/	10/		10/	10/	10/	10/	10/	10/	10/	10/	5.5 V					
	t_{ACS}		65	10/	10/	10/	10/	10/	10/	10/	10/		10/	10/	10/	10/	10/	10/	10/	10/	4.5 V	MAICH				
	t_{ACS2}		66	10/	10/	10/	10/	10/	10/	10/	10/		10/	10/	10/	10/	10/	10/	10/	10/	4.5 V	MAICH				
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = 125°C and T_C = -55°C.																									
9 T_C = 25°C	t_{AA}	Fig. 7, 8	67	11/	11/	11/	11/	11/	11/	11/	11/	GND	11/	11/	11/	11/	11/	11/	11/	11/	4.5 V	GALPA* and GALRISH	All I/O		70	ns
	t_{AA}		68																		5.5 V				70	
	t_{ACS1}		69																		4.5 V				70	
	t_{ACS1}		70																		5.5 V				70	
	t_{ACS2}		71																		4.5 V				80	
	t_{ACS2}		72																		5.5 V				80	
	t_{WC}		73																		4.5 V			70		
	t_{WC}		74																		5.5 V			70		
	t_{WP}		75																		4.5 V			50		
	t_{WP}		76																		5.5 V			50		
	t_{WR}		77																		4.5 V			5		
	t_{WR}		78																		5.5 V			5		

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Algorithms 1/	Measured terminal	Test limits Min	Test limits Max	Unit
				A_6	A_5	A_4	A_3	A_0	A_1	A_2	$\overline{CS}$	V_{SS}	$\overline{WE}$	I/O_4	I/O_3	I/O_2	I/O_1	A_9	A_8	A_7	V_{CC}					
T_C = 25°C	t_{DW}	Fig. 7, 8	79	11/	11/	11/	11/	11/	11/	11/	11/	GND	11/	11/	11/	11/	11/	11/	11/	11/	4.5 V	GALPAT and GALRESH	All I/O	25		ns
	t_{DW}		80																		5.5 V			25		
	t_{DH}		81																		4.5 V			0		
	t_{DH}		82																		4.5 V			0		
10	Same tests, terminal conditions, and limits as subgroup 9, except T_C = 125°C.																									
11	Same tests, terminal conditions, and limits as subgroup 9, except T_C = -55°C.																									

1/ See appendix for description of algorithms.

2/ I_{CC} = 180 mA.

3/ V_{IL} = GND, V_{IH} = 3.0 V, [illegible] ns/loop max, CS = high, [illegible] performed once at 125°C, and V_{CC} = 7.0 V min.

4/ V_{IL} = 0.8 V and V_{IH} = 2.0 V.

5/ Algorithm has 80 ns where chip is deselected between the write.

6/ V_{IL} = GND, V_{IH} = 3.0 V, and all address setup times are at minimums.

7/ V_{IL} = GND, V_{IH} = 3.0 V, and all write pulse timing are at minimums.

8/ V_{IL} = GND, V_{IH} = 3.0 V, and all address ending timing are at minimums.

9/ V_{IL} = GND, V_{IH} = 3.0 V, and t_{AA} is measured at minimum timing.

10/ V_{IL} = GND, V_{IH} = 3.0 V, t_{ACS1} and t_{ACS2} are measured at minimum timing.

11/ V_{IL} = 0.8 V, V_{IH} = 2.0 V, and all parameters are measured at minimum timing.

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6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V_{CC}	Supply voltage
V_{SS}	Common or reference voltage node
CS	Chip-selection, input
D_{IN}	Data input
D_{OUT}	Data output
A_0 thru A_{11}	Address input
WE	Read or write input
I_{OHZ}	High-impedance-state high output current
I_{OLZ}	High-impedance-state low output current
I_{CC}	Supply current from V_{CC} supply
t_{RC}	Read cycle time
t_{AS}	Address set-up time
t_{WP}	Write pulse width
t_{WR}	Write recovery time
t_{DW}	Data valid to end of write

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t_{DH}	Data hold time
t_{OH}	Output hold time from address change
t_{AA}	Address access time
t_{ACS}	Chip selection to output valid
t_{WC}	Write cycle time
t_{CW}	Chip selection to end of write
t_{AW}	Address valid to end of write
t_{LZ}	Chip selection to output in low impedance
t_{HZ}	Chip deselection to output in high impedance
t_{PU}	Chip selection to power up time
t_{PD}	Chip deselection to power down time
t_{WZ}	Write enabled to output in high impedance
t_{OW}	Output active from end of write
I_{LO}	Output leakage current
T_C	Case temperature
T_A	Ambient temperature

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	2147, 9147-85
02	2114, 9114B
03	2147H, 9147-70
04	2114A, 9114D
05	2147H-3, 9147-55
06	2148H, 9148-70
07	2147H-2, 9147-45

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6.7 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surfaces.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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APPENDIX

FUNCTIONAL ALGORITHMS

Functional algorithms are test patterns which define the exact sequence of tests used to verify proper operation of a random access memory (RAM). Each algorithm serves a specified purpose for the testing of the device.

10. FUNCTIONAL PATTERNS.

10.1 Pattern 1.

CKBD.

a. Write a checkerboard pattern into memory (0 in address 0) from address 0 to N.

b. When the $\overline{CS}$ off test is performed, attemp to write the complement pattern into cell memory with the device not selected.

c. Read checkerboard pattern in the memory.

10.2 Pattern 2.

$\overline{CKBD}$. Same as CKBD only with data complemented.

10.3 Pattern 3.

MARCH.

a. Write test word into every location.

b. The addressing is then scanned from location "0" to location "N".

c. At each address, the test word is read and a complemented test word is written back into the same location.

d. The addressing is then scanned in reverse from location "N" to location "0".

e. At each address, the complemented test word is read and the test word is written back in.

10.4 Pattern 4.

GALPAT. This program will test all bits in the array. The addressing and interaction between bits for ac performance. The memory is initialized by writing a field of "1" and then a field of "0" into the cell memory.

a. Write a "1" in word location 0 (reference location).

b. Word 0 is read.

c. Word 1 is read.

d. Word 0 is read.

e. Word 2 is read.

f. Word 0 is read.

g. The reading procedure continues back and forth between word 0 and the next higher number word until word 4095(-1) or 1023(-2) is reached. Then increment to the next word which becomes the reference location and then step a through g again until all the words in the memory are used at least once as a reference.

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10.5 Pattern 5.

Diagonal GALRE3H (with row column ping pong read CC II). This pattern will test all bits in the array for writing interaction for switching performance.

a. Initialize the memory by writing a field of 0's.

b. Perform the following read write sequence moving the test bit along the diagonal of the memory; and reading only the row and column of the test bit in ping ping fashion:

RO = Read "0" WI = Write "1" etc.

	STEP								
	1	2	3	4	5	6	7	8	9
BACKGROUND BIT		RO		RO		RO		RO	
TEST BIT	RO		WI		RI		WO		RO

c. Reinitialize the memory by writing a field of 1's.

d. Perform the following read write sequence moving the test bit along the diagonal of the memory; and reading only the row and column of the test bit in ping pong fashion:

	STEP								
	1	2	3	4	5	6	7	8	9
BACKGROUND BIT		RI		RI		RI		RI	
TEST BIT	RI		WO		RO		WI		RI

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